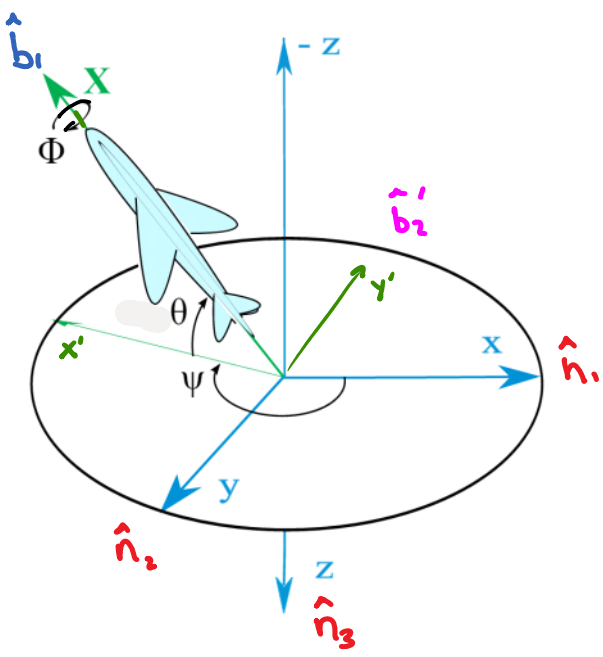
Euler Angles

- Most commonly used set of attitude coordinates
- 3 sequential finite rotations about body axes can uniquely determine the attitude of B $\{\hat{b}\}$ frame relative to N $\{\hat{n}\}$ frame
- Order (sequence) about which body axes we rotate matters! ie (3-2-1) $\neq$ (1-2-3)
- Because only 3 rotations (angles) are needed, this is a "minimal" set ... good ... but expect singularities



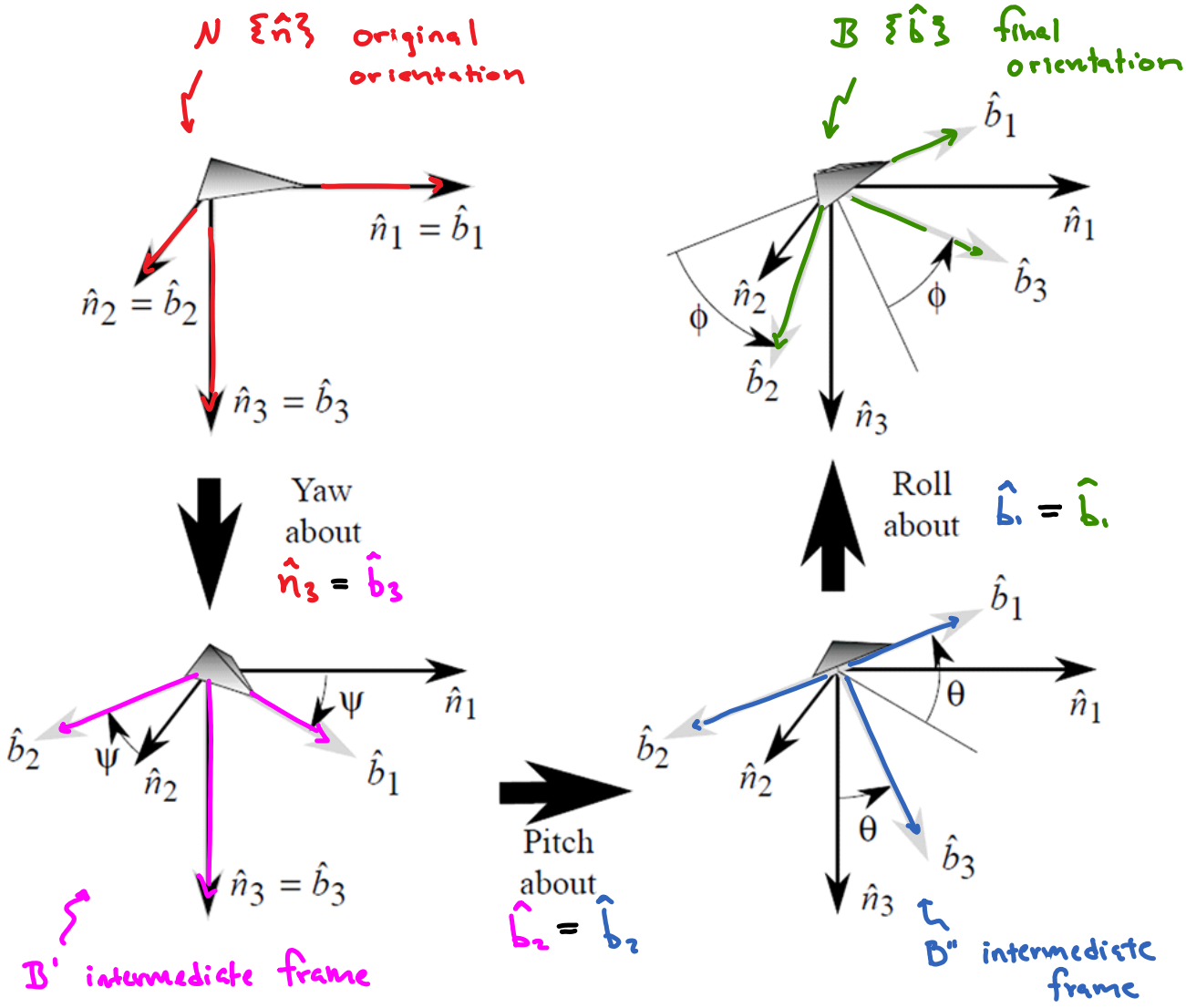
Yaw Pitch Roll (3-2-1 Euler Angles)

- First we consider the most common sequence of finite intrinsic body rotations:

① "Yaw" (ψ) about $\hat{n}_3$ (= initial $\hat{b}_3$)

② "Pitch" (θ) about $\hat{b}_2'$ (intermediate $\hat{b}_2$)

③ "Roll" (ϕ) about $\hat{b}_1$ (final $\hat{b}_1$)



- We can now at any instant in time define $\psi(t)$, $\theta(t)$, $\phi(t)$ such that their sequence, beginning from N, will define the attitude of B relative to N

Visualizing Tool For 3-2-1 Euler Angles

Geometric Singularity! If we pitch $\theta = \pm 90°$ then the $\hat{b}_1$ and $\hat{n}_3$ axes are lined up and yaw (ψ) and roll (ϕ) are not unique! This is physically related to "gimbal lock" phenomenon

EXAMPLE: Take an object and prove the following sequence is <u>not</u> unique:

yaw : 0° → pitch : -90° → roll : 45°

yaw : 45° → pitch : -90° → roll : 0°

ψ, ϕ not unique

... $\psi + \phi$ is unique

https://en.wikipedia.org/wiki/Gimbal_lock

Gimbal Lock Animation

Connecting to DCM:

- The Euler angles can be seen as 3 sequential planar rotations of $\{\tilde{b}\}$ (transformations) with corresponding 2D transformation matrices

$$[BB''] = \mathbb{T}_1(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}$$

$$[B''B'] = \mathbb{T}_2(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$[B'N] = \mathbb{T}_3(\psi) = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Recall the "composition rule": multiplying DCM corresponding to sequential rotations results in a new DCM that describes the attitude between the initial frame N and current frame B

3-2-1 DCM:

$$\underbrace{[C]}_{[BN]} = [BB''][B''B'][B'N] = \mathbb{T}_1(\phi)\mathbb{T}_2(\theta)\mathbb{T}_3(\psi)$$

$$= \begin{bmatrix} c\theta\, c\psi & c\theta\, s\psi & -s\theta \\ s\phi\, s\theta\, c\psi - c\phi\, s\psi & s\phi\, s\theta\, s\psi + c\phi\, c\psi & s\phi\, c\theta \\ c\phi\, s\theta\, c\psi + s\phi\, s\psi & c\phi\, s\theta\, s\psi - s\phi\, c\psi & c\phi\, c\theta \end{bmatrix}$$

with inverse relations:

$$\boxed{\begin{aligned} \psi &= \tan^{-1}\left(\frac{C_{12}}{C_{11}}\right) \\ \theta &= -\sin^{-1}\left(C_{13}\right) \\ \phi &= \tan^{-1}\left(\frac{C_{23}}{C_{33}}\right) \end{aligned}}$$

Careful!

- $\tan^{-1}$ returns 2 results ... use "atan2" in Matlab
- $-90^\circ \leq \theta \leq 90^\circ$

3-2-1 Euler Angle Kinematic Differential Eqn

Angular velocity ($\vec{\omega}_{B/N}$):

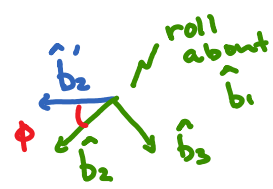
$$\vec{\omega} = \omega_1 \hat{b}_1 + \omega_2 \hat{b}_2 + \omega_3 \hat{b}_3$$

In terms of Euler angle rates:

$$\vec{\omega} = \dot{\psi}\, \hat{n}_3 + \dot{\theta}\, \hat{b}_2' + \dot{\phi}\, \hat{b}_1$$

Transforming $\hat{b}_2'$ and $\hat{n}_3$:

$$\hat{b}_2' = \cos\phi\, \hat{b}_2 - \sin\phi\, \hat{b}_3$$



$$\hat{n}_3 = -\sin\theta\, \hat{b}_1 + \sin\phi\cos\theta\, \hat{b}_2 + \cos\phi\cos\theta\, \hat{b}_3$$

recall $\{\hat{n}_i\} = [C]^T \{\hat{b}_i\}$

Relating ${}^B\vec{\omega}(t)$ to $\psi(t)$, $\theta(t)$, $\phi(t)$

$$\boxed{{}^B\begin{Bmatrix}\omega_1\\ \omega_2\\ \omega_3\end{Bmatrix} = \begin{bmatrix}-\sin\theta & 0 & 1\\ \sin\phi\cos\theta & \cos\phi & 0\\ \cos\phi\cos\theta & -\sin\phi & 0\end{bmatrix}\begin{Bmatrix}\dot{\psi}\\ \dot{\theta}\\ \dot{\phi}\end{Bmatrix}}$$

or after inverting the above relations:

$$\boxed{\begin{Bmatrix}\dot{\psi}\\ \dot{\theta}\\ \dot{\phi}\end{Bmatrix} = \frac{1}{\cos\theta}\begin{bmatrix}0 & \sin\phi & \cos\phi\\ 0 & \cos\phi\cos\theta & -\sin\phi\cos\theta\\ \cos\theta & \sin\phi\sin\theta & \cos\phi\sin\theta\end{bmatrix}{}^B\begin{Bmatrix}\omega_1\\ \omega_2\\ \omega_3\end{Bmatrix}}$$

$$= [B(\psi,\theta,\phi)]\,{}^B\vec{\omega}$$

"3-2-1 Euler Kinematic Diff Eq"

- Given initial conditions and ${}^B\vec{\omega}(t)$ these eqns can now be integrated to solve for $\psi(t)$, $\theta(t)$, and $\phi(t)$

- Note that these eqns become singular due to $(1/\cos\theta)$ term for $\theta = \pm 90°$, corresponds to geometric singularity

EXAMPLE: Consider two space-crafts B and F with attitudes relative to some N frame.

$\psi_B = 30°$, $\theta_B = -45°$, $\phi_B = 60°$ and $\psi_F = 10°$, $\theta_F = 25°$, $\phi_F = -15°$

Determine the attitude of B relative to F in terms of 3-2-1 Euler Angles.

Form DCM from eqn $[C(\psi,\theta,\phi)]$

$$[BN] = \begin{bmatrix} 0.612 & 0.353 & 0.707 \\ -0.780 & 0.127 & 0.612 \\ 0.127 & -0.927 & 0.354 \end{bmatrix}$$

$$[FN] = \begin{bmatrix} 0.893 & 0.157 & -0.423 \\ -0.275 & 0.732 & -0.235 \\ 0.357 & 0.326 & 0.875 \end{bmatrix}$$

Composition rule for DCM of B w.r.t. F:

$$[BF] = \underbrace{[BN][FN]^T}_{[NF]} = \begin{bmatrix} 0.303 & -0.005 & 0.952 \\ -0.935 & 0.190 & 0.299 \\ -0.182 & -0.982 & 0.053 \end{bmatrix}$$

Inverse relations for $\psi_{B/F}$, $\theta_{B/F}$, $\phi_{B/F}$:

$$\psi_{B/F} = \tan^{-1}\left(\frac{BF_{12}}{BF_{11}}\right) = -0.93^\circ \quad //$$

$$\theta_{B/F} = -\sin^{-1}(BF_{13}) = -72.33^\circ \quad //$$

$$\phi_{B/F} = \tan^{-1}\left(\frac{BF_{23}}{BF_{33}}\right) = 79.96^\circ \quad //$$

Interesting! Yaw of B wrt F negligible.

Note: in order to determine $\psi/\theta/\phi$ we had to convert Euler angles to DCM and then use the DCM composition rule ...

use the DCM composition rule ...
no simple similar rule exists for
Euler angles ☹